JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn, A Chance to Grow"

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Department of Computer Science & Engineering

B. Tech. in Computer Science & Engineering VII SEM/ 4th Year

Syllabus: 2026-27

Education to Eternity

॥ज्ञानम् सर्वार्थ साधनम्॥

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Program Name: B. Tech. Computer Science & Engineering **Program Code: CS**

Semester VII

Sr. No	Category of Subject	Course Code	Course Name	Teaching Scheme			Evaluation Scheme				Credit	Teaching Mode
				L	T/A	P	CA	MSE	ESE	Total		
1	PCC	CS7T001	Data Science	2	0	0	20	20	60	100	2	Class Board /PPT
2	PCC	CS7T002	Cyber Security & Cryptography	2	0	0	20	20	60	100	3	Class Board /PPT
3	PCC	CS7T003	Natural Language Processing	3	0	0	20	20	60	100	3	Class Board /PPT
4	ELC	CS7R001	Research Methodology	0	0	2	60	-	40	100	2	Class Board /PPT
5	PEC	CS7E006	PEC-VI	3	0	0	20	20	60	100	3	Class Board /PPT
6	PEC	CS7E007	PEC-VII	3	0	0	20	20	60	100	3	Class Board /PPT
7	MDM	CS7M007	MDM-VII	2	0	0	20	20	60	100	2	Class Board /PPT
8	PCC	CS7L001	Cyber Security & Cryptography Lab	0	0	2	60	-	40	100	1	PCs
9	PCC	CS7L002	Natural Language Processing Lab	0	0	2	60	-	40	100	1	PCs
10	ELC	CS7P001	Project Phase-I	0	0	2	60	-	40	100	1	Project
11	ELC	CS7I002	Internship/ OJT-II	0	0	0	60	-	40	100	1	Internship
			Total	**15**	**0**	**8**	**420**	**120**	**560**	**1100**	**22**	

Version of the Scheme	**JD/ACD/CSE/2023-27/4**	**Signatures of all BOS members**
Year of release of the scheme document:	**2025-26**	

Secretary BOS

Chairman BOS

Deputy Dean (Acd)

Dean (Acd)

Dean (IQAC)

Chairman (Acd.Council)

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Professional Elective Courses (PEC) List

Sr. No.	Subject Code	CSE Domain				
		PEC Category	**Database**	**Computing Domain**	**Computer Vision Domain**	**Systems**
1	**CS5E006**	**PEC-VI**	Transition & Concurrency Control	Quantum Computing	Advance Computer Vision	Reinforcement Learning
2	**CS5E007**	**PEC-VII**	Advanced Database	Affective Computing	Pattern Recognition	Randomized Algorithm
3	**CS7M007**	**MDM-VII**	Generative AI			

Version of the Scheme	JD/ACD/CSE/2025-29/1	Signatures of all BOS members
Year of release of the scheme document:	**2025-26**	

Secretary BOS

Chairman BOS

Deputy Dean (Acd)

Dean (Acd)

Dean (IQAC)

Chairman (Acd.Council)

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7T001	**Data Science**	2	0	0	2

Prerequisites for the course	
1.	Basics Knowledge of Data Science and algorithm.
2.	Understand sorting, searching and hashing algorithms.

Prior Reading Material/useful links	
1	https://www.google.co.in/books/edition/What_Is_Data_Science/-OQ2q5JqOdEC?hl=en&gbpv=1&dq=Data+Science&printsec=frontcover
2	https://www.google.co.in/books/edition/Data_Science_for_Business/4ZctAAAAQBAJ?hl=en&gbpv=1&dq=Data+Science&printsec=frontcover
3	https://www.google.co.in/books/edition/Getting_Started_with_Data_Science/b4YxCwAAQBAJ?hl=en&gbpv=1&dq=Data+Science&printsec=frontcover

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students shall be able to Build the fundamentals of data science.
2	CO2	Students shall be able to Apply Data Collection and Data Pre-processing Strategies.
3	CO3	Students shall be able to Compare and choose data visualization method for effective visualization of data
4	CO4	Students shall be able to Implement regression models, model evaluation and validation
5	CO5	Students shall be able to Test Multiple Parameters by using Grid Search

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Data Science:** What is Data Science, importance of data science, Big data and data Science, The current Scenario, Industry Perspective Types of Data: Structured vs. Unstructured Data, Quantitative vs. Categorical Data, Big Data vs. Little Data, Data science process, Role Data Scientist.	6

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Unit II	**Data Collection and Data Pre-Processing:** Data Collection Strategies, Data Pre-Processing Overview, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization.	6
Unit III	**Exploratory Data Analytics :** Descriptive Statistics, Mean, Standard Deviation, Skewness and Kurtosis, Box Plots, Pivot Table, Heat Map, Correlation Statistics.	6
Unit IV	**Model Development :** Simple and Multiple Regression, Model Evaluation using Visualization, Residual Plot, Distribution Plot, Polynomial Regression and Pipelines, Measures for In-sample Evaluation, Prediction and Decision Making, Feature Engineering	6
Unit V	**Model Evaluation :** Generalization Error, Out-of-Sample Evaluation Metrics, Cross Validation, Overfitting, Under Fitting and Model Selection, Prediction by using Ridge Regression, Testing Multiple Parameters by using Grid Search	6

Text Books	
1.	Mitchell, Tom. M., "Data Science", McGraw-Hill Education, 1st Edition, May 2013.
2.	Segaran, Toby. "Programming Collective Intelligence- Building Smart Web 2.0 Applications", O'Reilly Media, August 2007.
3.	Miroslav, Kubat. "An Introduction to Data Science", Springer Publishing.

Reference Books	
1.	Bishop, C. M., "Data Science and AI", Springer Publishing.
2.	Conway, Drew and White, John Myles, "Data Analytics", O'Reilly Media, February 2012.

Useful Links	
1.	http://nptel.ac.in/courses/106105031/lecture by Dr. Debdeep Mukhopadhyay IIT Kharagpur
2.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-computer-system engineering-spring-2009/video-lectures/ lecture by Prof. Robert Morris and Prof. Samuel Madden MIT.
3.	https://www.google.co.in/books/edition/Applied_Cryptography_and_Network_Securit/srsyEAAAQBAJ?hl=en&gbpv=1&dq=cyber+security+%26+cryptography+notes&printsec=frontcover

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7T002	Cyber Security & Cryptography	3	0	0	3

Prerequisites for the course	
1.	Basics Knowledge of Cyber Security.
2.	Understand various Authentication algorithms

Prior Reading Material/useful links	
1.	https://blog.rsisecurity.com/what-is-cryptography-in-cyber-security/
2.	https://www.tutorialspoint.com/what-is-the-difference-between-cryptography-and-cyber-security
3.	https://www.vssut.ac.in/lecture_notes/lecture1428550736.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Understand basic concepts of Cyber security.
2	CO2	Student will able to Apply security principles to system design and Symmetric Encryption algorithms to provide confidentiality
3	CO3	Student will able to Compare and apply various authentication Techniques and different cryptographic operations of public key cryptography.
4	CO4	Student will able to Evaluate and Communicate the human role in security systems with an emphasis on ethics, social engineering vulnerabilities.
5	CO5	Student will able to Select and apply appropriate Intrusion detection and prevention techniques and to examine various security algorithms to Interpret security incidents

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Cyber Security:** Overview of cyber security, Internet Governance-Challenges and constraints, Cyber threats:-Cyber Warfare-Cyber Crime-Cyber terrorism ,Cyber Espionage, Need for	6

	comprehensive cyber security policy, need for nodal authority, Cyber security regulations, Roles of international law.	
Unit II	**Cryptography and Block Ciphers principles:** Introduction, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security, Substitution and Transposition techniques, Symmetric and Asymmetric key cryptography, Steganography, Cryptographic independent dimensions. Cryptanalytic attack and brute force attack, Symmetric key Ciphers: Block Cipher principles, DES.	6
Unit III	**Public Key Cryptosystems and Authentication Requirements:** Principles of public key cryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, introductory idea of Elliptic curve cryptography. Cryptographic Hash Functions: Message Authentication, Secure Hash Algorithm (SHA-512) Message authentication codes: Authentication requirements, Digital signature	6
Unit IV	**Key Management, Distribution and Cyber Security Vulnerabilities** Distribution of Public Keys, Kerberos, X.509 Authentication Service, PGP, SSL, IPSEC. Cyber Security Vulnerabilities-Overview, vulnerabilities in software, System administration, Complex Network Architectures, Weak Authentication, Poor Cyber Security Awareness. Cyber Security Safeguards- Overview, Access control, Audit, Authentication, Ethical Hacking.	6
Unit V	**Securing Web Application Services, Servers and cyber forensics:** Basic security for HTTP Applications and Services, Basic Security for SOAP Services, Identity Management and Web Services, Security Considerations, Challenges, Intrusion detection and Prevention Techniques, System Integrity Validation, Honey pots, password management. Introduction to Cyber Forensics.	6

Text Books	
1.	William Stallings, "Crpyptography and Network security Principles and Practices", Pearson/PHI.
2.	Wade Trappe, Lawrence C Washington, " Introduction to Cryptography with coding theory", Pearson.
3.	J. Katz and Y. Lindell, Introduction to Modern Cryptography, CRC press, 2008.
4.	Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing – Prentice Hall of India.

Reference Books	
1.	Golreich O, Foundations of Cryptography, Vol.1., Cambridge University Press, 2004
2.	Menezes, et.al, Handbook of Applied Cryptography, CRC Press, 2004.
3.	Introduction to Cryptography and Network Security By Behrouz A. Forouzan · 2008, McGraw-Hill Higher Education
4.	Cyber Security Cryptography and Machine Learning Fourth International Symposium, CSCML 2020, Be'er Sheva, Israel, July 2–3, 2020, Springer International Publishing.

Useful Links	
1.	http://nptel.ac.in/courses/106105031/lecture by Dr. Debdeep Mukhopadhyay IIT Kharagpur
2.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-computer-system engineering-spring-2009/video-lectures/ lecture by Prof. Robert Morris and Prof. Samuel Madden MIT.
3.	https://www.google.co.in/books/edition/Applied_Cryptography_and_Network_Securit/srsyEAAAQBAJ?hl=en&gbpv=1&dq=cyber+security+%26+cryptography+notes&printsec=frontcover
4.	https://www.google.co.in/books/edition/Applied_Cryptography_and_Network_Securit/OrBI6J5wHewC?hl=en&gbpv=1&dq=cyber+security+%26+cryptography+notes&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7T003	Natural Language Processing	3	0	0	3

Prerequisites for the course	
1.	Basics of Neural Networks and Machine Learning and Process.
2.	Study the Character Encoding, Word Segmentation, Sentence Segmentation.

Prior Reading Material/useful links	
1.	https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligence_natural_language_processing.htm
2.	https://www.cl.cam.ac.uk/teaching/2002/NatLangProc/nlp1-4.pdf
3.	https://www.geeksforgeeks.org/introduction-to-natural-language-processing/

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Apply the principles and Process of Human Languages such as English and other Indian Languages using computers.
2	CO2	Student will able to Realize semantics and pragmatics of English language for text processing.
3	CO3	Student will able to Create CORPUS linguistics based on digestive approach (Text Corpus method)
4	CO4	Student will able to Check a current methods for statistical approaches to machine translation.
5	CO5	Student will able to Perform POS tagging for a given natural language and Select a suitable language modelling technique based on the structure of the language.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to NLP** - Various stages of NLP –The Ambiguity of Language: Why NLP Is DifficultParts of Speech: Nouns and Pronouns, Words: Determiners and adjectives, verbs, Phrase Structure. Statistics	7

	Essential Information Theory : Entropy, perplexity, The relation to language, Cross entropy.	
Unit II	Character Encoding, Word Segmentation, Sentence Segmentation, Introduction to Corpora, Corpora Analysis. Inflectional and Derivation Morphology, Morphological analysis and generation using Finite State Automata and Finite State transducer.	7
Unit III	N gram models, Smoothing, Part of speech tagging, Hidden Markov models, Viterbi algorithm, Forward - backward algorithm, EM training, Models for Named Entity Recognition, Neural Language Models - Recurrent Neural Networks and Long Short term Memory networks	7
Unit IV	**Methodological Preliminaries, Supervised Disambiguation:** Bayesian classification, An information theoretic approach, Dictionary-Based Disambiguation: Disambiguation based on sense, The saurus based disambiguation, Disambiguation based on translations in a second-language corpus.	7
Unit V	**Markov Model:** Hidden Markov model, Fundamentals, Probability of properties, Parameter estimation, Variants, Multiple input observation. The Information Sources in Tagging: Markov model taggers, Viterbi algorithm, Applying HMMs to POS tagging, Applications of Tagging	8

Text Books	
1.	Christopher D. Manning and Hinrich Schutze, " Foundations of Natural Language Processing" , 6 th Edition, The MIT Press Cambridge, Massachusetts London, England, 2003
2.	Daniel Jurafsky and James H. Martin "Speech and Language Processing", 3rd edition, Prentice Hall, 2009.
3.	NitinIndurkhya, Fred J. Damerau "Handbook of Natural Language Processing", Second Edition, CRC Press, 2010.
4.	James Allen "Natural Language Understanding", Pearson Publication 8th Edition. 2012..

Reference Books	
1.	Chris Manning and HinrichSchütze, "Foundations of Statistical Natural Language Processing", 2nd edition, MITPress Cambridge, MA, 2003.
2.	Hobson lane, Cole Howard, Hannes Hapke, "Natural language processing in action" MANNING Publications, 2019.

3.	Neural Network Methods in Natural Language Processing By Yoav Goldberg · 2017
4.	Deep Learning for Natural Language Processing Develop Deep Learning Models for your Natural Language Problems By Jason Brownlee · 2017

Useful Links

1	https://www.google.co.in/books/edition/Introduction_to_Natural_Language_Process/72yuDwAAQBAJ?hl=en&gbpv=1&dq=natural+language+processing+Course+Objectives&printsec=frontcover
2	https://www.cs.mcgill.ca/~jcheung/teaching/fall-2015/comp599/comp599-outline.pdf
3	https://luddy.iupui.edu/courses/info-b443/
4	https://www.cl.cam.ac.uk/teaching/2002/NatLangProc/nlp1-4.pdf

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7R001	Research Methodology	2	0	0	2

Prerequisites for the course	
1.	Understand and use basic data analysis techniques.
2.	Study the basic data collection methods with emphasis on secondary and survey research.

Prior Reading Material/useful links	
1.	https://mrcet.com/downloads/digital_notes/CSE/Mtech/I%20Year/RESEARCH%20METHODLOGY.pdf
2.	https://ccsuniversity.ac.in/bridgelibrary/pdf/MPhil%20Stats%20Research%20Methodology-Part1.pdf
3.	https://www.drnishikantjha.com/papersCollection/Research%20Methodology%20.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will able to Identify a research problem stated in a study
2	CO2	Students will able to Obtain skills to identify a business problem/ need, translate it into a research question, and design an appropriate way to answer it.
3	CO3	Students will able to Develop skills to design a research project and collect data.
4	CO4	Students will able to Develop skills to critically evaluate the quality of other researchers‘ findings and the process used to obtain them.
5	CO5	Students will able to Identify the overall process of designing a research study from its inception to its report.

Syllabus:

Course Contents		Hours
Unit I	**Introduction:** Meaning, Objectives, Research process, Methods and Methodology, Criteria of good research, Review of literatures: Primary source, Secondary source, Identifying gap areas from	9

	literature review, Searching e- resources, using search engines, Searching data base.	
Unit II	**Types of Research:** Pure research, applied research, Exploratory Research, Descriptive research, Diagnostic research, Quantitative and Qualitative research etc.	9
Unit III	**Research Sampling and Design: Sampling of data:** Concept of sampling, Probability sampling techniques , Non probability sampling techniques , Sampling error, Research Design: Meaning, Need, Types of research design-Exploratory Research Design, components of research design and features of good research design,	10
Unit IV	**Methods, Collection and Analysis of Data:** Types of data, Methods of data collection- Interview Method, Mailing Method, Observation Method, Survey Method etc.; Primary and secondary sources of data, Sampling- meaning and methods, Classification and Tabulation, Graphical presentation, Application of computer in research data analysis.	10
Unit V	**Presentation of Research: Citation Styles-** APA, MLA etc., Research ethics and Plagiarism, Indexing of journal and research output, Report writing steps in report writing, layout of report writing, reference and bibliography.	9

Text Books	
1.	Research Methodology, Methods and Techniques by C.R Kothari, 2nd Edition.
2.	Sinha, S.C. and Dhiman, A.K., 2002. Research Methodology, Ess Ess Publications. 2 volumes. 4.
3.	The Science of Education Research, Eurasia Publishing House, New Delhi by George J. (1964),
4.	Advanced focus Group Research, Sage Publication, India Ltd, New Delhi by Fern Edward F. (2001)

Reference Books	
1.	Research Methodology in Management, Himalaya Publishing House, New Delhi by Michael V.P.
2.	Trochim, W.M.K., 2005. Research Methods: the concise knowledge base, Atomic Dog Publishing. 270p.
3.	Research Methodology, A Handbook for Beginners, By Pagadala Suganda Devi · 2017

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E006A	PEC-VI (Transaction & Concurrency Control)	3	0	0	3

Prerequisites for the course	
1.	Fundamentals of Database Management Systems
2.	Basic understanding of Operating Systems
3.	Knowledge of data structures and algorithms

Prior Reading Material/useful links	
1.	Basic DBMS concepts
2.	Relational Algebra and SQL basics
3.	NPTEL: DBMS - Prof. Sudarshan, IIT Bombay

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will able to understand the concept of transactions and the need for concurrency control in DBMS.
2	CO2	Student will able to Analyse and resolve problems caused by concurrent execution of transactions.
3	CO3	Student will able to Apply different concurrency control techniques in transaction processing.
4	CO4	Student will able to implement recovery techniques to maintain database consistency.
5	CO5	Student will able to Examine the effects of concurrency and recovery in distributed and modern databases.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Transactions:** ACID Properties, Transaction states, System crash and its effects, Schedules and serializability, Conflict serializability and view serializability, Precedence graphs.	8

Unit II	**Concurrency Control – Issues and Techniques**: Concurrency problems: lost update, temporary update, incorrect summary, uncommitted data, Lock-based protocols: binary locks, shared/exclusive locks, Two-Phase Locking (2PL) protocol, Strict 2PL, Deadlocks and starvation, Wait-Die and Wound-Wait protocols, Deadlock prevention, detection, and recovery.	**6**
Unit III	**Time-Based and Optimistic Concurrency Control:** Timestamp ordering protocol, Thomas' write rule, Multiversion concurrency control (MVCC), Validation-based (optimistic) protocol, Comparison of concurrency control methods.	**8**
Unit IV	**Types of failures:** system crash, transaction failure, media failure, Recovery concepts, Log-based recovery: undo, redo, and undo/redo logging, Checkpoints and fuzzy checkpointing, Shadow paging, ARIES recovery algorithm (overview).	**8**
Unit V	Distributed transaction management, Two-Phase Commit Protocol (2PC), Nested transactions, Concurrency in distributed databases, Modern NoSQL databases and concurrency (brief overview), Snapshot isolation in modern databases.	**6**

Text Books	
1.	Abraham Silberschatz, Henry F. Korth, S. Sudarshan, "Database System Concepts", 7th Edition, McGraw-Hill Education.
2.	Jim Gray and Andreas Reuter, "Transaction Processing: Concepts and Techniques", Morgan Kaufmann.
3.	M. Tamer Özsu and Patrick Valduriez, "Distributed Database Systems", 2nd Edition, Pearson.

Reference Books	
1.	Ramez Elmasri and Shamkant B. Navathe, "Fundamentals of Database Systems", 7th Edition, Pearson Education.
2.	Philip A. Bernstein and Eric Newcomer, "Principles of Transaction Processing", 2nd Edition, Morgan Kaufmann.

Useful Links	
1.	DBMS Full Course by Codebasics (Beginner Friendly)
2.	NPTEL: Database Management System (Prof. P. Dasgupta, IIT Kharagpur)

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E006B	PEC-VI (Quantum Computing)	3	0	0	3

Prerequisites for the course	
1.	Basic of Quantum Computation.
2.	Fundamental of Cryptography.
3.	Understanding different Quantum Algorithms.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/introduction-quantum-computing/
2.	https://www.hpe.com/us/en/what-is/quantum-computing.html
3.	https://thequantuminsider.com/introduction-to-quantum-computing/

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Students will able to understand Quantum Computation.
2	CO2	Students will gain an understanding the Quantum mechanics as applied in Quantum computing.
3	CO3	Students will be able to describe the Troubleshoot and improve deep learning models.
4	CO4	Students will be able to Optimization and Generalization in neural networks.
5	CO5	Students will be Implement deep learning algorithms, understand neural networks and traverse the layers of data abstraction which will empower the student to understand data more precisely.

Syllabus:

	Course Contents	Hours
Unit I	**Introduction to Quantum Computation:** Quantum bits, B loch sphere representation of a qubit, multiple qubits.	6
Unit II	**Background Mathematics and Physics:** Hilber space, Probabilities and measurements, entanglement, density operators and correlation,	6

	basics of quantum mechanics, Measurements in bases other than computational basis.	
Unit III	**Quantum Mechanics**: Measurements in bases other than computational basis. 083 Quantum Circuits: single qubit gates, multiple qubit gates, design of quantum circuits.	**6**
Unit IV	**Quantum Information and Cryptography:** Comparison between classical and quantum information theory. Bell states, Quantum teleportation. Quantum Cryptography, no cloning theorem.	**6**
Unit V	**Quantum Algorithms:** Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor factorization, Grover search. Noise and error correction: Graph states and codes, Quantum error correction, fault-tolerant computation.	**12**

Text Books	
1.	Nielsen M. A ., Quantum Computation and Quantum Information, Cambridge University Press.2002
2.	Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol.I : Basic Concepts, Vol I I: Basic Tools an d Special Topics, World Scientific.2004
3.	Pittenger A. O., An Introduction to Quantum Computing Algorithms.2000

Reference Books	
1.	Quantum computing explained, David McMahon, Wiley-interscience, John Wiley & Sons, Inc. Publication 2008
2.	Quantum computation and quantum information, Michael A. Nielsen and Isaac L. Chuang, Cambridge University Press 2010
3.	Introduction to Quantum Mechanics, 2nd Edition, David J. Griffiths, Prentice Hall New Jersey 1995

Useful Links	
1.	https://www.reddit.com/r/QuantumComputing/?rdt=35140
2.	https://www.quantum-inspire.com/kbase/introduction-to-quantum-computing/

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E006C	PEC-VI (Advanced Computer Vision)	3	0	0	3

Prerequisites for the course

1.	Understanding on detailed models of image formation.
2.	Examine various clustering algorithms

Prior Reading Material/useful links

1.	https://www.studocu.com/in/document/amrita-vishwa-vidyapeetham/computer-vision/computer-vision-lecture-notes-all/23202964
2.	http://eecs.northwestern.edu/~yingwu/teaching/EECS432/Notes/vision.pdf
3.	https://www.mccormick.northwestern.edu/electrical-computer/academics/courses/descriptions/432.html

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Appreciate the detailed models of image formation.
2	CO2	Student will able to Analyse the techniques for image feature detection and matching.
3	CO3	Student will able to Apply various algorithms for pattern recognition.
4	CO4	Student will able to Examine various clustering algorithms.
5	CO5	Student will able to Analyze structural pattern recognition and feature extraction techniques

Syllabus:

Course Contents		Hours
Unit I	**Image formation and Image model-** Components of a vision system- Cameras- camera model and camera calibration- Radiometry- Light in space- Light in surface - Sources, shadows and shading.	7
Unit II	**Multiple images-**The Geometry of multiple views- Stereopsis- Affine structure from motion- Elements of Affine Geometry Affine	7

	structure and motion from two images- Affine structure and motion from multiple images- From Affine to Euclidean images.	
Unit III	**High level vision-** Geometric methods- Model based vision- Obtaining hypothesis by pose consistency, pose clustering and using Invariants, Verification.	7
Unit IV	**Introduction to pattern and classification,** supervised and unsupervised learning, Clustering Vs classification, Bayesian Decision Theory- Minimum error rate classification Classifiers, discriminant functions, decision surfaces- The normal density and discriminant-functions for the Normal density.	8
Unit V	Linear discriminant based classifiers and tree classifiers Linear discriminant function based classifiers- Perceptron- Minimum Mean Squared Error (MME) method, Support Vector machine, Decision Trees: CART, ID3.	7

Text Books	
1.	Bernd Jahne and Horst HauBecker, Computer vision and Applications, Academic press, 2000.
2.	David A. Forsyth & Jean Ponce, Computer vision – A Modern Approach, Prentice Hall, 2002.
3.	C. M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
4.	R. O. Duda, P. E. Hart and D. G. Stork, Pattern Classification, John Wiley, 2001.

Reference Books	
1.	Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Second Edition, Cambridge University Press, 2004.
2.	S. Theodoridis and K. Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009.
3.	Computer Vision and Image Processing Fundamentals and Applications By Manas Kamal Bhuyan · 2019
4.	Advanced Topics in Computer Vision, 2013, Springer London

Useful Links	
1.	https://www.google.co.in/books/edition/Concise_Computer_Vision/ZCu8BAAAQBAJ?hl=en&gbpv=1&dq=advanced+computer+vision+notes&printsec=frontcover
2.	https://faculty.ucmerced.edu/mcarreira-perpinan/teaching/ee589/lecture-notes.pdf
3.	http://vision.stanford.edu/teaching/cs131_fall1718/files/cs131-class-notes.pdf

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E006D	PEC-VI (Reinforcement Learning)	3	0	0	3

Prerequisites for the course	
1.	Basic of mathematical foundations of reinforcement learning.
2.	Examine various reinforcement learning techniques to real-world problems.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/what-is-reinforcement-learning/
2.	https://www.techtarget.com/searchenterpriseai/definition/reinforcement-learning
3.	https://aws.amazon.com/what-is/reinforcement-learning/

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will able to understand the basic concepts of reinforcement learning
2	CO2	Student will able to analyse dynamic programming algorithms, Monte-Carlo learning, and Temporal-difference learning.
3	CO3	Student will able to describe and apply Incremental Metho21DS, Batch Metho21DS, and Deep Q-Learning metho21DS.
4	CO4	Student will able to develop strategies for balancing exploration and exploitation in reinforcement learning problems.
5	CO5	Student will able to analyse and design multi-agent reinforcement learning systems..

Syllabus:

Course Contents		Hours
Unit I	**Probability Primer**-Brush up of Probability concepts, Introduction to Reinforcement Learning(RL): Recent Advancements and Highlights, The RL Problem and Overall Lan21DScape. Markov Decision Process (MDP): Markov Process, Markov Reward Process, Markov Decision Process and Bellman Equations, Partially Observable MDPs.	**7**

Unit II	**Dynamic Programming:** Policy Evaluation, Value Iteration, Policy Iteration, DP Extensions and Convergence using Contraction Mapping. Monte-Carlo (MC) Learning, Temporal-Difference (TD) Learning, TD(λ) and Eligibility Traces. On-Policy MC Control, OnPolicy TD Learning, Off-Policy Learning, SARSA and Q-Learning.	**7**
Unit III	**Value Function Approximation**: Incremental Metho21DS (Linear and Gradient based), Batch Metho21DS (Least Square based).Deep RL: Deep Q-Learning, Deep Q-Networks and Experience Replay.	**7**
Unit IV	**Exploration and Exploitation (Bandits):** Exploration Principles (Greedy, Optimistic, Probabilistic, Informative), Multi-arm Bandits, Contextual Bandits and Upper Confidence Boun21DS (UCB), Upper Confidence Reinforcement Learning (UCRL).	**8**
Unit V	**Multi-Agent RL:** Cooperative vs. Competitive Settings, Mixed Setting, Game-Theoretic Formulation, MARL Algorithms Applications and Case Studies.	**7**

Text Books	
1.	Reinforcement Learning: An Introduction, Richard S. Sutton, 2nd edition, MIT Press, 2020
2.	Reinforcement Learning, second edition: An Introduction Richard S. Sutton, Andrew G. Barto · 2018
3.	Reinforcement Learning, Phil Winder Ph.D. · 2020
4.	Deep Reinforcement Learning Hands-On, Maxim Lapan · 2024

Reference Books	
1.	Grokking Deep Reinforcement Learning, Miguel Morales · 2020
2.	Algorithms for Reinforcement Learning, Csaba Szepesvári · 2022
3.	Deep Reinforcement Learning: Frontiers of Artificial, Mohit Sewak · 2020

Useful Links	
1.	https://www.ibm.com/think/topics/reinforcement-learning#:~:text=Reinforcement%20learning%20(RL)%20is%20a,instruction%20by%20a%20human%20user.
2.	https://in.mathworks.com/discovery/reinforcement-learning.html
3.	https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E007A	PEC-VII (Advance Database)	3	0	0	3

Prerequisites for the course	
1.	Database Management Systems
2.	Database Management Systems (Lab)

Prior Reading Material/useful links	
1.	https://ecomputernotes.com/database-system/adv-database
2.	https://www.business.rutgers.edu/sites/default/files/documents/phd-syllabus-advanced-database-systems.pdf
3.	https://learn.saylor.org/course/view.php?id=91

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Students will able to Apply the basic concepts of Advance Database Systems and Applications.
2	CO2	Students will able to Use of the Advance database and construct queries using SQL in database creation and interaction
3	CO3	Students will able to Describe Apache Cassandra InterfacesAnalyse and techniques of Cassandra Command Line Interface.
4	CO4	Students will able to Apply the basic concepts of Basic operations with MongoDB shell.
5	CO5	Students will able to implement integration of security and recovery in database systems;

Syllabus:

Course Contents		Hours
Unit I	**Introduction to ADB:** Introduction to Advanced Database , Comparision of DBMS & ADBMS, DBMS Advanced Features and Distributes Database (Query Processing and Evaluation, Transaction Management and Recovery, Database Security and Authorisation, Distributed Databases), Types of ADBS (Network database Systems, Object-Oriented Database Systems, Hierarchical Database Systems)	6

Unit II	**NoSQL database concepts:** Differentiate SQL and NoSQL databases. Types of NoSQL databases, NoSQL data modeling, Benefits of NoSQL, comparision between SQL and NoSQL database system. Use NoSQL database to solve given queries.NoSQL using MongoDB: Introduction to MongoDB Shell, Running the MongoDB shell, MongoDB client, Basic operations with MongoDB shell,	**8**
Unit III	**Big Data:** Overview of Big Data and NoSQL Database: The 3 Vs. of Big Data, Data Evolution ,Features of Big Data ,Big Data-Use Cases , Big Data Analytics, Traditional Technology vs. Big Data Technology,ApacheHadoop , HDFS, Map Reduce , NoSQL Databases, Approaches to NoSQL Databases-Types	**7**
Unit IV	**Apache Cassandra:** Introduction to Apache Cassandra: Characteristics, History of Cassandra, Features of Cassandra ,When is Cassandra Used ? , Simple Cassandra Program, Cassandra Command Line Interface, Advantages of Cassandra, Limitations of Cassandra.	**7**
Unit V	**Apache Cassandra Interfaces:** Cassandra supports Cassandra Query Language or CQL, DDL and DML Statements ,DML Statements – COPY Apache Cassandra Interfaces :Cassandra Interfaces , Cassandra Command Line Interface ,Cqlsh Options ,Cqlsh Commands ,Cqlsh Shell Commands , Querying Cassandra	**8**

Text Books	
1.	Mastering Apache Cassandra - Second Edition by NishantNeeraj (Author)
2.	Henry Korth, Abraham Silberschatz& S. Sudarshan, Database System Concepts, McGraw-Hill Publication, 6th Edition, 2011

Reference Books	
1.	Wiederhold, Database Design, McGraw-Hill Publication, 2nd Edition, 1983.
2.	Advanced Database Systems, By Nabil R. Adam, Bharat K. Bhargava · 1993, Springer

Useful Links	
1.	https://www.youtube.com/watch?v=hKljaVcCMgg
2.	https://www.youtube.com/watch?v=poEfLYH9W2M

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	**CS7E007B**	**PEC-VII (Affective Computing)**	**3**	**0**	**0**	**3**

Prerequisites for the course	
1.	Understanding of intelligent systems and AI concepts.
2.	Familiarity with rule-based systems and machine learning basics.

Prior Reading Material/useful links	
1.	https://www.datacamp.com/blog/what-is-affective-computing
2.	https://www.datacamp.com/blog/what-is-affective-computing
3.	https://spj.science.org/doi/10.34133/icomputing.0076

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Students will able to Understand the fundamentals of affective computing and emotional models.
2	CO2	Students will gain and apply discrete and dimensional models of emotions.
3	CO3	Students will be able to Implement emotion detection systems using machine learning and deep learning frameworks.
4	CO4	Students will be able to Create emotionally expressive avatars, robots, and affective dialogue agents.
5	CO5	Students will be Analyse and evaluate affective computing applications with ethical considerations.

Syllabus:

	Course Contents	Hours
Unit I	**Introduction to Affective Computing:** Definition and Scope of Affective Computing, Historical Background and Evolution, Emotions: Theories and Models, Role of Emotion in Human Cognition and Decision Making, Applications of Affective Computing in Real World.	**8**

Unit II	Emotion Representation and Modeling: Taxonomy of Emotions Emotion Annotation Schemes, Affective State Models: OCC Model, PAD Model Multimodal Emotion Representation Emotion Ontologies and Standards (e.g., EmotionML)	**8**
Unit III	Emotion Detection Techniques: **Facial Expression Recognition** (FER): FACS, CNNs, and OpenFace, **Speech Emotion Recognition**: Prosodic, Spectral, and Linguistic Features, **Text-Based Emotion Detection**: NLP, Sentiment Analysis, Transformer Models, **Physiological Signal Analysis**: ECG, EEG, Skin Conductance, Multimodal Fusion Approaches.	**8**
Unit IV	Emotion Synthesis and Generation: Emotion Generation in Avatars and Robots, Affective Dialogue Systems and Chatbots, Affective Speech and Text Generation, Emotional Gesture and Facial Animation, Deepfake and Ethical Issues in Emotion Synthesis.	**6**
Unit V	Applications, Challenges, and Future Directions: Applications, Ethical, Privacy, and Social Implications, Bias and Fairness in Emotion AI, Future Trends: Emotion in XR, Empathic AI, Human-Robot Relationships	**6**

Text Books	
1.	"Affective Computing", Rosalind W. Picard, MIT Press, **ISBN**: 9780262661157
2.	"Emotion-Oriented Systems: The Humaine Handbook", Paolo Petta, Catherine Pelachaud, Roddy Cowie, Springer, **ISBN**: 9783540795712
3.	**"The Oxford Handbook of Affective Computing",** Rafael A. Calvo, Sidney D'Mello, Jonathan Gratch, Arvid Kappas, Oxford University Press.

Reference Books	
1.	"Affective Computing and Sentiment Analysis", K.R. Venugopal, L.M. Patnaik, Springer, **ISBN**: 9789811551006
2.	**"Emotion Measurement" (2nd Edition),** Herbert L. Meiselman, Woodhead Publishing, **ISBN**: 9780081017432
3.	**"Affective Computing and Intelligent Interaction"** Sidney D'Mello, Art Graesser, Björn Schuller, Jean-Claude Martin, *Springer*, **ISBN**: 9783642452070

Useful Links	
1.	https://onlinecourses.nptel.ac.in/noc23_cs36/preview
2.	https://www.datacamp.com/blog/what-is-affective-computing

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E007C	PEC-VII (Pattern Recognition)	3	0	0	3

Prerequisites for the course	
1.	A basic of Neural Network and fuzzy logic.
2.	A Basic of Python programming.

Prior Reading Material/useful links	
1.	https://onlinecourses.nptel.ac.in/noc19_ee56/preview
2.	https://www.isical.ac.in/~k.ramachandra/PR_Course.htm
3.	https://www.geeksforgeeks.org/pattern-recognition-introduction/

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will able to Summarize the various techniques involved in pattern recognition.
2	CO2	Students will able to Categorize the various pattern recognition techniques into supervised and unsupervised.
3	CO3	Students will able to Illustrate the artificial neural network-based pattern recognition.
4	CO4	Students will able to Discuss the applications of pattern recognition in various applications
5	CO5	Students will able to analyze Hidden Markov model and Support Vector Machine

Syllabus:

Course Contents		Hours
Unit I	**Pattern Classifier:** Overview of Pattern recognition, Discriminant functions, supervised learning, parametric estimation, Maximum Likelihood Estimation.	7
Unit II	**Bayes Classifier:** Bayesian parameter Estimation, Problems with Bayes approach, Pattern classification by distance functions, Minimum	7

	distance pattern classifier.	
Unit III	**Clustering:** Clustering for unsupervised learning and classification Clustering concept, CMeans algorithm, Hierarchical clustering, Graph theoretic approach to pattern Clustering, Validity of Clusters.	7
Unit IV	**Feature Extraction and Structural Pattern Recognition:** KL Transforms, Features election through functional approximation, Binary selection, Elements of formal grammars, Syntactic description, stochastic grammars, Structural representation**.**	7
Unit V	**Hidden Markov model and Support Vector Machine:** State machine, Hidden Markov model, Training, Classification, Support vector machine, Feature Selection. Recent Advances: Fuzzy logic, Fuzzy Pattern Classifier, Pattern classification using genetic algorithms.	8

Text Books	
1.	M. Narasimha Murthy and V. Susheela Devi, "Pattern Recognition", Springer 2011
2.	S. Theodoridis and K. Koutroumbas, "Pattern Recognition", 4th Ed., Academic Press, 2009.
3.	Robert J. Schalkoff, "Pattern Recognition Statistical, Structural and Neural Approaches", John Wiley and Sons Inc., New York, 1992
4.	C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.

Reference Books	
1.	The Black Swan: The Impact of the Highly Improbable (Hardcover)
2.	Pattern Recognition and Machine Learning (Information Science and Statistics)
3.	Introduction to Statistical Pattern Recognition (Hardcover)
4.	How to Create a Mind: The Secret of Human Thought Revealed (Hardcover)

Useful Links	
1.	https://onlinecourses.nptel.ac.in/noc19_ee56/preview
2.	http://www.cse.msu.edu/~cse802/
3.	https://cedar.buffalo.edu/~srihari/CSE555/
4.	https://www.geeksforgeeks.org/pattern-recognition-introduction/

Education to Eternity

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7E007D	PEC-VII (Randomized Algorithms)	3	0	0	3

Prerequisites for the course	
1.	Understand the principles of random signals.
2.	Study the basics of Design and Analysis of Algorithm.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/randomized-algorithms/
2.	https://ocw.mit.edu/courses/6-856j-randomized-algorithms-fall-2002/
3.	http://theory.stanford.edu/people/pragh/amstalk.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students have the basics of probability, events and random experiments.
2	CO2	They can analyze that the random variable is always a numerical quantity.
3	CO3	Students can use the multiple random variables and relate through examples to real problems.
4	CO4	Students can have the concept of random processes in both deterministic and non deterministic types.
5	CO5	Students can Use the Power density spectrum and its properties and the types of noise

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Randomized Algorithms:** Review of Basic Probability, Polynomial Identity Testing, Schwartz - Zippel Lemma, Reduction from Perfect Bipartite Matching to PIT, Randomized Quick sort, Markov, Chebyshev, and Chernoff bounds, Tossing coins, coupon collector problem, birthday paradox, Balls and bins, Two point sampling.	7

Unit II	**Randomized rounding:** Multi-commodity flow, Introduction to Markov chain, randomized algorithm for 2SAT, stationary distribution, Irreducible and aperiodic Markov chain, fundamental theorem of Markov chain (statement only), coupling and Random walk	7
Unit III	**Metropolis Algorithm**: Mixing time of Random Walk on Cycles, Proof of the fundamental Theorem of Markov chains, Finishing proof of the fundamental Theorem of Markov chains, hitting time, commute time, cover time, Monte Carlo Method, FPRAS for DNF Counting, FPRAS for Independent Set Counting using Monte Carlo Method	7
Unit IV	**Introduction to Probabilistic Methods:** Probabilistic method of expectation, alteration; Lovasz Local Lemma and its application, Method of Conditional Expectation for De-randomization, Overview of path coupling.	7
Unit V	**Introduction to Universal Hash Family**: Perfect Hashing, Cuckoo Hashing, Bloom Filter, Count Min Sketch, Construction of Universal Hash Family, Locality Sensitive Hashing (LSH), Nearest Neighbor Search (NNS), Point Location in Equal Balls (PLEB), Johnson Lindenstrauss Lemma Sub-Gaussian Random Variables	8

Text Books	
1.	Randomized Algorithms: Rajeev Motwani, Prabhakar Raghavan, Cambridge University Press.
2.	Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis by Eli Upfal and Michael Mitzenmacher
3.	Computational Geometry: Algorithms and Applications, by Mark de Berg, Otfried Cheong, Marc van Kreveld, and Mark Overmars, 3rd edition, Springer-Verlag, 2008.
4.	Algorithmic and Analysis Techniques in Property Testing, by Dana Ron. Found. Trends Theor. Comput. Sci. 5, 2 (February 2010), 73-205.

Reference Books	
1.	Randomized Algorithms, By Rajeev Motwani · 2001, Cambridge University Press
2.	Probability and Computing Randomized Algorithms and Probabilistic Analysis By Michael Mitzenmacher, Eli Upfal · 2005
3.	Towards Dynamic Randomized Algorithms in Computational Geometry By Monique Teillaud · 1993

4.	Probability and Computing Randomization and Probabilistic Techniques in Algorithms and Data Analysis By Michael Mitzenmacher, Eli Upfal · 2017

Useful Links	
1.	https://www.google.co.in/books/edition/Randomized_Algorithms/QKVY4mDivBEC?hl=en&gbpv=1&dq=Randomized+Algorithms&printsec=frontcover
2.	https://www.google.co.in/books/edition/Randomized_Algorithms_for_Analysis_and_C/KyGmPavw7OMC?hl=en&gbpv=1&dq=Randomized+Algorithms&printsec=frontcover
3.	https://www.google.co.in/books/edition/Design_and_Analysis_of_Randomized_Algori/XARmmrB86UEC?hl=en&gbpv=1&dq=Randomized+Algorithms&printsec=frontcover
4.	https://www.google.co.in/books/edition/Randomized_Algorithms_in_Automatic_Contr/sK0qBAAAQBAJ?hl=en&gbpv=1&dq=Randomized+Algorithms&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7M007	MDM-VII (Generative AI)	2	0	0	2

Prerequisites for the course	
1.	Artificial Intelligence
2.	Machine Learning and Deep Learning Fundamentals
3.	Basic understanding of Neural Networks and Python Programming

Prior Reading Material/useful links	
1.	https://www.tensorflow.org/tutorials/generative
2.	https://huggingface.co/docs

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Students will be able to Understand the theoretical foundation and evolution of generative modeling techniques.
2	CO2	Students will be able to Apply Variation Auto encoders, GANs, and Transformer-based models for data generation.
3	CO3	Students will be able to Analyze performance metrics, challenges, and limitations of generative models.
4	CO4	Students will be able to Implement generative AI models for text, image, and multimodal applications using deep learning frameworks.
5	CO5	Students will be able Evaluate ethical considerations, bias mitigation, and societal impacts of Generative AI.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Generative AI** Definition, history, and scope of Generative AI. Overview of discriminative vs. generative models. Generative AI applications in text, image, audio, and video. Probability theory and latent variable models.	6
Unit II	**Generative Modeling Techniques** Auto encoders (AE), Variational Auto encoders (VAE) – theory, loss functions, parameterization trick. GANs – architecture, generator, discriminator, loss functions, training challenges, and evaluation metrics.	6

Unit III	**Advanced Generative Models** Conditional GANs, Cycle GANs, Style GANs. Diffusion Models – denoising diffusion probabilistic models (DDPM). Normalizing Flows. Comparison of generative approaches.	6
Unit IV	**Transformers and Multimodal Generation** Attention mechanism, Transformer architecture, Large Language Models (LLMs), Vision Transformers (ViT). Text-to-image models (e.g., DALL·E, Stable Diffusion). Speech and video generation techniques.	6
Unit V	**Applications, Ethics, and Case Studies:** Applications in art, content creation, healthcare, education, and simulation. Responsible AI principles, data privacy, copyright, deepfake detection, and bias mitigation strategies. Emerging trends: Retrieval-Augmented Generation (RAG), Agents, and OpenAI ecosystem.	6

Text Books	
1.	David Foster-Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play by O'Reilly Media
2.	Ian Goodfellow, Yoshua Bengio, Aaron Courville-Deep Learning by MIT Press
Reference Books	
1.	Venkat Ankam-Hands-On Generative AI with Python by Packt Publishing
2.	Peter Welinder et al-Building Generative AI Applications with LLMs by O'Reilly Media

Useful Links	
1.	https://www.tutorialspoint.com/gen-ai/index.htm
2.	OpenAI Research Blog
3.	https://www.geeksforgeeks.org/artificial-intelligence/what-is-generative-ai/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7L001	Cyber Security & Cryptography Lab	0	0	2	1

Prerequisites for the course	
1.	Study the basics of Cyber Security & Cryptography.
2.	Understand various applications of cryptography and security issues.

Prior Reading Material/useful links	
1.	https://cse29-iiith.vlabs.ac.in/
2.	http://www.anuraghyd.ac.in/cse/wp-content/uploads/sites/10/NS-CRYPTO-LAB-Final11.pdf
3.	https://sriindu.ac.in/wp-content/uploads/2023/02/R18CSE41L1-Crytography-Network-Security-Lab.pdf

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Student will able to Analyse and resolve security issues in networks and computer systems to secure an IT infrastructure.
2	CO2	Student will able to Examine various Security algorithms.
3	CO3	Student will able to Know the methods of conventional encryption
4	CO4	Student will able to Understand various applications of cryptography and security issues practically.
5	CO5	Student will able to Understand the concepts of public key encryption and number theory.

List of Practical's:

Course Contents		Hours
1	Introduction of Cyber Security & Cryptography Lab.	2
2	Encrypt the message "The key is hidden under the door pad" with the encryption key "information" using play fare cipher.	2

3	Employ column transposition techniques to encrypt the message "attack postponed until two am", the column transposition encryption key is 3 4 2 1 5 6 7 decrypt the cipher text to get back the plain text message	2
4	Generate random number generator using Blum Blum shub and encrypt message using bitwise XOR operation.	2
5	Perform the encryption and decryption of message using RC4 algorithm.	2
6	Encrypting and Decrypting Messages Using Morse Code use site https://morsecode.world/international/translator.html	2
7	WAP for RSA Algorithm using HTML and JavaScript.	2
8	WAP for Diffie-Hellman Key Exchange algorithm.	2
9	Implementation of the message digest of a text using the SHA-1 algorithm	2
10	Implement the SIGNATURE SCHEME - Digital Signature Standard.	2

Note: 2 Practical based on Content Beyond Syllabus

Text Books	
1.	The Network Security Test Lab, A Step-by-Step Guide, By Michael Gregg · 2015
2.	Introduction to Cryptography and Network Security, By Behrouz A. Forouzan · 2008
3.	Lab Manual to Accompany Access Control, Authentication, and Public Key Infrastructure, By Bill Ballad, Tricia Ballad, Erin K. Banks · 2011

Reference Books	
1.	Introduction to Computer and Network Security, Navigating Shades of Gray By Richard R. Brooks · 2013
2.	Hands-On Information Security Lab Manual By Michael E. Whitman, Herbert J. Mattord · 2010
3.	Fundamentals of Cryptography Introducing Mathematical and Algorithmic Foundations, By Duncan Buell · 2021

Useful Links	
1.	https://sriindu.ac.in/wp-content/uploads/2023/02/R18CSE41L1-Crytography-Network-Security-Lab.pdf
2.	https://www.csa.iisc.ac.in/~cris/about.html
3.	https://www.vvitengineering.com/lab/odd/CS6711-Security-Lab-Manual.pdf
4.	https://www.gvpce.ac.in/syllabi/MTech15-16/cyber-security/nsclab.pdf

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7L002	Natural Language Processing Lab	0	0	2	1

Prerequisites for the course	
1.	Basic Concepts of Data Structures & Algorithms.
2.	Programming Knowledge.

Prior Reading Material/useful links	
1.	https://web.stanford.edu/class/cs224n/
2.	https://huggingface.co/learn/nlp-course/
3.	https://www.coursera.org/specializations/natural-language-processing

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Understand approaches to syntax and semantics in NLP.
2	CO2	Demonstrate approaches to discourse, generation, dialogue and summarization within NLP.
3	CO3	Apply current methods for statistical approaches to machine translation.
4	CO4	Recognize the significance of pragmatics for natural language understanding.

List of Practical's:

Course Contents		Hours
0	Study of Natural Language Processing lab.	2
1	Implement sentiment analysis technique for classifying the data in to positive, negative or neutral class	2
2	Use of NLP technique for text summarization	2
3	Implement simple machine translation from one language to another	2
4	Implement a code for aspect mining and topic modelling.	2
5	Preprocessing of text (Tokenization, Filtration, Script Validation, Stop Word Removal, Stemming)	2

6	TO implement POS tagging.	2
7	To implement chunking.	2
8	To implement name entity recognition.	2
9	Perform the word generator on Virtual Lab .	2
10	Mini Project based on NLP Application	2

Note: 2 Practical based on Content Beyond Syllabus

Text Books	
1.	**Speech and Language Processing** **By:** Daniel Jurafsky & James H. Martin **Edition:** 3rd Edition (Draft available online)**Description:** Comprehensive and widely used textbook covering linguistics, statistical models, and deep learning for NLP.
2.	**Foundations of Statistical Natural Language Processing By:** Christopher D. Manning & Hinrich Schütze **Description:** A classic book on statistical approaches in NLP — essential for understanding early NLP systems and probabilistic models. **Publisher:** MIT Press **Level:** Intermediate to advanced

Reference Books	
1.	Deep Learning for Natural Language Processing By: Palash Goyal, Sumit Pandey, and Karan Jain Description: Covers modern deep learning techniques applied to NLP like word embeddings, RNNs, and Transformers. Level: Intermediate to advanced
2.	**Natural Language Understanding By:** James Allen **Description:** One of the earlier foundational books focusing on AI-based language understanding .**Publisher:** Pearso n**Level:** Theoretical, suited for early NLP/AI researchers

Useful Links	
1.	**ACL Anthology (NLP Research Papers Archive)** https://aclanthology.org/

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7P001	Project Phase I	0	0	2	1

Prerequisites for the course	
1.	Study the new technologies or improving existing ones.
2.	Develop new computational models or simulations that help advance their research.

Prior Reading Material/useful links	
1.	https://www.youtube.com/watch?v=p8e5ZEpRmx0
2.	https://www.youtube.com/watch?v=if_z7pMA85g
3.	https://www.youtube.com/watch?v=V5yv5TNpiLE

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will able to creation of new technologies or innovations.
2	CO2	Students will able to address complex problems or challenges
3	CO3	Students will able to develop new computational models or simulations that help advance their research.
4	CO4	Students will able to create new products or services that can be commercialized.
5	CO5	Students will able to impact various aspects of society, from improving productivity and efficiency to enhancing user experience and addressing social challenges.

The project should enable the students to combine the theoretical and practical concepts studied in his/her academics. The project work should enable the students to exhibit their ability to work in a team, develop planning and execute skills and perform analyzing and trouble shooting of their respective problem chosen for the project. The students should be able to write technical report, understand the importance of teamwork and group task. The students will get knowledge about literature survey, problem definition, its solution, and method of calculation, trouble shooting, costing, application and scope for future development.

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Project work

The project work is an implementation of learned technology. The knowledge gained by studying various subjects separately supposed to utilize as a single task. A group of 03/04 students will have to work on assigned work. The topic could be a product design, specific equipment, live industrial problem etc. The project work involves experimental/theoretical/computational work. It is expected to do necessary literature survey by referring current journals belonging to Information Technology reference books and internet. After finalization of project, requisites like equipments, data, tools etc. should be arranged.

Project Activity

The project groups should interact with guide, who in turn advises the group to carry various activities regarding project work on individual and group basis. The group should discuss the progress every week in the project hours and follow further advice of the guide to continue progress. Guide should closely monitor the work and help the students from time to time. The guide should also maintain a record of continuous assessment of project work progress on weekly basis.

Phase I

1) Submission of Review Paper or Research Paper or Conference and Copyright Details.
2) Submission of project/problem abstract containing problem in brief, requirements, broad area, applications, approximate expenditure if required etc.
3) Problem definition in detail.
4) Literature survey.
5) Requirement analysis.
6) System analysis (Draw DFD up to level 2, at least).
7) System design, Coding/Implementation (50 to 80%).

Text Books	
1.	Planning and Implementing Your Final Year Project — with Success! By Mikael Berndtsson, Jörgen Hansson, B. Olsson, Björn Lundell · 2013
2.	Computer Science Project Work Principles and Pragmatics By Sally Fincher, Marian Petre, Martyn Clark · 2001
3.	Thesis Projects, A Guide for Students in Computer Science and Information SystemsBy Mikael Berndtsson · 2008

Reference Books	
1.	SOFTWARE ENGINEERING PROJECT MANAGEMENT By Bill Brykczynski, Richard D. Stutz · 2006

2.	Software Engineering Body of Knowledge By IEEE Computer Society · 2014
3.	Quality Software Project Management By Robert T. Futrell · 2002

Useful Links	
1.	https://www.google.co.in/books/edition/Project_Summaries/0ggTG_ya8acC?hl=en&gbpv=1&dq=major+project+objective+in+computer+science+engineering&pg=PA78&printsec=frontcover
2.	https://www.google.co.in/books/edition/Real_World_Software_Projects_for_Compute/X6seEAAAQBAJ?hl=en&gbpv=1&dq=major+project+objective+in+computer+science+engineering&printsec=frontcover
3.	https://www.logicraysacademy.com/blog/final-year-projects-for-cse/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7I002	Internship-II	0	0	0	1

Prerequisites for the course	
1.	Basics of Internet Programming like coding, designing, Testing.
2.	Fundamental Knowledge of Computing and Programming.

Prior Reading Material/useful links	
1.	https://www.indeed.com/career-advice/career-development/internship-report
2.	https://www.scribbr.com/academic-writing/internship-report/

Course Outcomes:

Sr. No	Course Outcome No	CO statement
1	CO1	Students will be able to Apply theoretical knowledge to real-world industrial or research problems.
2	CO2	Students will be able to Develop practical skills in tools, technologies, and professional practices.
3	CO3	Students will be able to Demonstrate project planning, execution, and documentation skills
4	CO4	Students will be able to Communicate work effectively through reports and presentations.
5	CO5	Students will be able to Gain insights into industrial workflows, team collaboration, and problem-solving.

Course Details:

Provides the student with an opportunity to gain knowledge and skills from a planned work experience in the student's chosen career field. In addition to meeting Core Learning Outcomes, jointly developed Specific Learning Outcomes are selected and evaluated by the Faculty Internship Advisor, Worksite Supervisor, and the student. Internship placements are directly related to the student's program of study and provide learning experiences not available in the classroom setting. Internships provide entry-level, career-related experience, and workplace competencies that employer's value when hiring new employees. Internships may also be used as an opportunity to explore career fields. Students must meet with the Internship

& Apprenticeship Coordinator prior to registering. The purpose of the Internship Program is to provide each student practical experience in a standard work environment. The Internship & Apprenticeship Coordinator and Faculty Internship Advisor will assist students in making the job a valuable and productive experience. Success in this job will help ensure development of skills necessary for a lasting and rewarding career in the future.

1. Internship Work:

- Duration: 1 Months- Full-time or 3Month-Online (8 hrs/day or as per institute norms)
- Work on a project assigned by the industry/organization under supervision.
- Apply relevant technologies, tools, and methodologies learned in coursework.
- Maintain daily/weekly work logs and document progress.

2. Progress Monitoring:

- Regular interaction with internal faculty guide and external industry supervisor.
- Monthly progress review meetings.
- Progress Report Submission
- Submit periodic progress reports (monthly or bi-monthly) to faculty mentor.
- Include project objectives, work completed, challenges, and next steps.

3. Final Report Submission:

- Prepare a comprehensive internship report covering
- Organization profile
- Internship objectives and scope
- Methodologies, tools, and technologies used
- Work carried out, results achieved, challenges faced
- Learning outcomes and future recommendations
- Format: As per institute guidelines (usually 40–60 pages, typed, A4, Times New Roman 12 pt).

4. Seminar / Presentation:

- Duration: 15–20 minutes presentation, followed by Q&A.
- Present internship work, results, and learning experiences to faculty and peers.
- Assessment based on clarity, content, presentation skills, and understanding of the project.

5. Assessment Components:

- Internship Work / Industry Supervisor Evaluation-40%
- Progress Report Submission-20%
- Final Report Submission-20%
- Seminar / Presentation -20%

Text Books	
1.	R. K. Sharma, Krishna Mohan"How to Prepare Internship Reports", 1st Edition, S. Chand Publishing, 2019
2.	R. R. Bowker-"Internship & Co-op Handbook: A Practical Guide to Career Success", 2nd Edition, 2016

Reference Books	
1.	R. C. Sharma"Professional Communication & Report Writing", 1st Edition, 2020
2.	Institute Guidelines / University Handbook – Most universities provide a structured internship report template and evaluation criteria.

Useful Links	
1.	https://www.scribbr.com/academic-writing/internship-report/
2.	https://www.indeed.com/career-advice/career-development/internship-report

Sr. No.	Course Code	Course Name	Subject In charge Name
1	CS7T001	Data Science	Prof. Sujata Helonde Prof. Dipali Pethe
2	CS7T002	Cyber Security & Cryptography	Prof. V. Karthikeyan
3	CS7T003	Natural Language Processing	Prof. Aachal Wani Prof. Komal Chavke
4	CS7R001	Research Methodology	Prof. Khushboo Bisen Prof. Sameer Mende
5	CS7E006B	PEC-VI (Quantum Computing)	Prof. Priyanka Patil
6	CS7E006D	PEC-VI (Reinforcement Learning)	Prof. Dipti Khobragade
7	CS7E007A	PEC-VII (Advanced Database)	Prof. Mittal Patne
8	CS7E007	PEC-VII (Affective Computing)	Prof. Maresh Shingade
9	CS7M007	MDM-VII (Generative AI)	Prof. Ragini Khobragade Prof. Karishma Bobde
10	CS7L001	Cyber Security & Cryptography Lab	Ms. Priyanka Barai
11	CS7L002	Natural Language Processing Lab	Ms. Pratiksha Kanhekar
12	CS7P001	Project Phase-I	Prof. Anuja Ghasad
13	CS7I002	Internship/ OJT-II	Mr. Sandip Zade

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Secretary BOS, CSE,
JDCOEM, Nagpur

Chairman BOS CSE,
JDCOEM, Nagpur

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BOS (CSE)
JDCOEM, Nagpur